

Tarek Mohamed Salah

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Summary

Frontend-focused Software Engineer with full-stack range across React, TypeScript, Java/Spring Boot, and Node-based backends (NestJS, Express). Recent Communications and Electronics Engineering graduate (Alexandria University, GPA 3.42/4.0, Excellent) who served as Technical Project Lead on EduVerse, a full-stack academic platform spanning web, mobile, and AI integrations, coordinating a 6-person team across frontend, backend, mobile, and AI workstreams, and previously led a 3-person team building Ajarly, a production-ready rental platform. Built three full-stack web platforms end to end, including a flight-booking system delivered on a 10-day client deadline.

Technical Skills

Frontend: React, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, shadcn/ui, Radix UI, React Hook Form, Zod, React Router, Responsive Design, Component-Based Architecture, State Management

Backend: Java, Spring Boot, Node.js, Express, NestJS, Python (FastAPI), RESTful APIs, JWT Authentication

Data & Infrastructure: MySQL, Docker, Supabase, Cloudinary, Socket.IO, Git & GitHub

Other Languages: Python, C, Embedded C, Assembly Language

Practices: Agile, Cross-Functional Team Leadership, API Integration, UI/UX Design, Version Control

Experience

Software Engineer Intern — Alex Eagles Aero Design

*Alexandria, Egypt
Sept 2024 – Present*

- Apply Python and embedded-systems knowledge to UAV control and obstacle-avoidance software as part of a 6-developer autonomous drone team (project detailed below).

Software Engineer Intern — ALX

*Remote
Feb 2023 – Feb 2024*

- Completed an intensive, project-based software engineering program covering C, Python, data structures and algorithms, Linux/Unix systems programming, shell scripting, and Git-based version control.
- Built and debugged backend and systems-level projects individually and with peers, developing a DevOps-oriented workflow for infrastructure and deployment basics.

React Frontend Web Developer Scholarship — Digital Egypt Pioneers Initiative (DEPI), Ministry of Communications and Information Technology

*Alexandria, Egypt (Hybrid)
Jun 2025 – Dec 2025*

- Completed a 6-month, cohort-based program (Cohort 3) building responsive, production-style applications with React, TypeScript, and modern UI/UX practices; recognized with a Certificate of Recognition as a top-performing participant.
- Delivered 3 freelance client projects during the program while working with Node.js, Express, Docker, and Kubernetes fundamentals.

Projects

EduVerse — Academic Management Platform (Graduation Project)

2025 – 2026

- Served as Technical Project Lead for a 6-person team, coordinating frontend, backend, mobile, and AI workstreams on a full-stack platform unifying learning, administration, and communication for 5 user roles: students, instructors, teaching assistants, administrators, and IT administrators.
- Owned frontend architecture, authentication, shared UI components, and the API layer; built role-based dashboards for each user role.
- Built and tested backend REST APIs for core academic workflows and designed the initial architecture for the Question Bank and Exam Generation modules.
- Integrated Google Drive, YouTube, and Python-based AI services into the platform; supported deployment and cross-module debugging.

- Extended the web-first platform to mobile with a Flutter app on the same backend.
- *Technologies:* React, TypeScript, NestJS, MySQL, Python (FastAPI), Docker, Flutter, Supabase, Socket.IO

Ajarly — Real Estate Rental Platform

2025

- Led a 3-person team building a rental platform connecting property owners, brokers, and renters in Egypt; directed frontend development and built major backend modules.
- Backend (Java + Spring Boot): implemented JWT authentication, property management, a booking system with availability and conflict handling, simulated payments with platform-fee and owner-share splits, a reviews system, and an analytics module with daily, monthly, and owner-level insights.
- Frontend (React + TypeScript): designed the complete UI/UX and built reusable components with shadcn/ui and Radix UI, implemented form validation with React Hook Form and Zod, and integrated dynamic search filters.
- Applied layered architecture, DTOs/mappers, centralized exception handling, and query indexing to keep the system production-ready rather than a typical course submission.
- *Technologies:* Java, Spring Boot, React, TypeScript, Tailwind CSS, JWT, Cloudinary

My Trip — Flight Booking and Management System

2025

- Built the entire frontend, solo, for a paid freelance flight-booking system covering search, booking and payment, e-ticketing, cancellations, and flight management, with dedicated portals for passengers, crew, front desk, and admin.
- Delivered production-ready and end to end on a 10-day client deadline, integrating REST APIs with Axios and managing application state and routing with React Router.
- *Technologies:* React.js, Tailwind CSS, Axios, React Router, Radix UI

Autonomous Drone System

Sept 2024 – Jun 2025

- Built and tested a drone control class using DroneKit-Python and MAVLink as part of a 6-developer team handling global path planning, SDK integration, and real-time mission visualization.
- *Technologies:* Python, DroneKit-Python, MAVLink, React

Education

Alexandria University

BS in Communication and Electronics Engineering

Alexandria, Egypt

Sept 2021 – Jul 2026

- GPA: 3.42/4.0 (Excellent)
- Core Coursework: Embedded Systems, Operating Systems, Data Structures & Algorithms, Assembly Language, Embedded C, Python, C, Project Management

Certifications & Training

- Certificate of Recognition, Top-Performing Student — Digital Egypt Pioneers Initiative (DEPI), Cohort 3, Ministry of Communications and Information Technology, Egypt
- Egypt FWD Web Development Challenger Track — Udacity, Issued Oct 2021 (Credential ID: AEVQGYKK)
- Egypt FWD Data Analysis Challenger Track — Udacity, Issued Sep 2022 (Credential ID: YUR35GTS)

Languages

Arabic (Native) | English (Full Professional)